MR_IL6R_CRC v51 JTM Revision Report

Manual GEO validation integration, Figure 6 candidate rerun, and supplementary source-data/code crosswalk

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1. Executive decision

v51 is scientifically stronger than v49 because the previous “GSE39582 not parsed” limitation has been resolved. GSE39582 now provides a reliable external tumour-normal comparison, and the expanded GEO validation supports a conservative, JTM-suitable interpretation: IL6R is not a robust tumour-intrinsic oncogene or standalone prognostic biomarker, but it is linked to immune/myeloid/inflammatory microenvironmental biology.

2. Key numeric updates

Tumour-normal validation (Supplementary Table S10)

cohort	n_tumour	n_normal	median_tumour	median_normal	mean_tumour	mean_normal	delta_median	wilcoxon_p
GSE39582	566	19	4.064	6.677	4.181	6.624	-2.613	3.94e-13

Stage-stratified validation (Supplementary Table S11)

cohort	n_stage_I_II	n_stage_III_IV	median_stage_I_II	median_stage_III_IV	delta_median_III_IV_minus_I_II	wilcoxon_p
GSE14333	138	152	2.980	2.836	-0.144	0.635
GSE17536	81	96	6.644	6.657	0.013	0.639
GSE17537	19	36	6.841	6.378	-0.463	0.261
GSE39582	309	270	4.106	4.044	-0.062	0.464

Survival validation (Supplementary Table S12)

cohort	endpoint	n	events	HR_per_1SD_IL6R	CI_low	CI_high	p_value
GSE17536	OS	177	73	0.897	0.709	1.135	0.366
GSE17537	OS	55	20	1.323	0.865	2.024	0.196
GSE39582	OS	573	190	0.993	0.867	1.138	0.922
GSE17536	DFS	145	36	1.003	0.723	1.391	0.987
GSE17537	DFS	42	6	0.567	0.208	1.546	0.267
GSE39582	DFS	536	147	0.928	0.786	1.096	0.380
GSE17536	DSS	177	55	0.923	0.704	1.209	0.560

3. Figure decisions

Figure 6: I generated a new v51 candidate from the v50 cell-level GSE146771 source table. It is cleaner and more directly supports the immune/myeloid IL6R-axis logic. Because it uses an expression/module-score source table rather than the full CellChat object, it should either replace the previous Figure 6 after local review or be used as a cleaner supplementary single-cell expression figure.

Supplementary Figure S2: Add this as the formal manual GEO validation figure. It directly addresses the prior independent-validation weakness and should be cited in Results and Discussion.

Figure source-data/code crosswalk: Add Supplementary Table S8 to improve reproducibility and satisfy reviewer expectations for computational transparency.

4. JTM compliance implications

JTM/Springer guidance states that multi-panel figures should be submitted as a single composite file, figure titles and legends should be in the manuscript rather than embedded only in the graphic, and datasets supporting conclusions should be available in repositories or supporting files. The v51 package follows this direction by adding composite figure files, standalone legends, source-data tables and a repository/Zenodo update note.

Remaining action: after local confirmation, create a new GitHub release (for example v1.1.0) and archive it in Zenodo to obtain a DOI. Then update the Data availability statement with the release URL/DOI.

5. Recommended manuscript logic

The strongest final narrative is: negative MR/colocalization and non-prognostic TCGA/GEO survival results argue against IL6R as a direct circulating-protein causal risk factor or standalone prognostic biomarker; lower tumour expression relative to normal/non-tumoral tissue argues against a simple tumour-intrinsic oncogene model; immune deconvolution, scRNA and support-layer analyses converge on IL6R as an immune/myeloid/inflammatory microenvironment marker.